

Satellite Structural Inspection Interferometric Synthetic Aperture Radar (InSAR) Based

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METHODOLOGY

1. Satellite SAR System with Resolution and Revisit Time

To select a suitable SAR system, technical specifications such as spatial resolution, frequency band, polarization, and revisit time were considered. Sentinel-1, operating at C-band (5.405 GHz), was selected due to:

Parameter	Sentinel-1 C-SAR (IW Mode)
Frequency Band	C-band (~5.405 GHz)
Revisit Time	12 days (single satellite)
Spatial Resolution	5 m (range) × 20 m (azimuth)
Coverage Mode	Interferometric Wide Swath (IW)

2. Retrieving Multi-Temporal SAR Imagery Over the Specified Period

Multi-temporal **Sentinel-1 SLC IW mode** data were downloaded based on:

- **Area of Interest (AOI):** A polygon covering the bridge and surrounding terrain.
- **Date range:** Multiple acquisitions were collected spanning several months to capture temporal variations.
- **Image acquisition conditions:** All images were selected with the same relative orbit, pass direction (ascending or descending), and polarization (typically VV/VH) to ensure consistency for interferometric analysis.

S1A_IW_SLC__1SDV_20250111T130240_20250111T130310_057397_0710AB_813D
S1A_IW_SLC__1SDV_20250123T130239_20250123T130309_057572_07179E_5431
S1A_IW_SLC__1SDV_20250204T130238_20250204T130308_057747_071E86_3FE6
S1A_IW_SLC__1SDV_20250216T130238_20250216T130308_057922_072592_C312
S1A_IW_SLC__1SDV_20250228T130238_20250228T130307_058097_072CB4_7A14
S1A_IW_SLC__1SDV_20250312T130238_20250312T130308_058272_0733CA_217F
S1A_IW_SLC__1SDV_20250324T130238_20250324T130308_058447_073AB5_B445
S1A_IW_SLC__1SDV_20250405T130238_20250405T130308_058622_0741CE_2B52
S1A_IW_SLC__1SDV_20250417T130238_20250417T130308_058797_0748F4_3FFD
S1A_IW_SLC__1SDV_20250429T130239_20250429T130309_058972_07501E_08D8
S1A_IW_SLC__1SDV_20250511T130238_20250511T130308_059147_0756BA_DE13
S1A_IW_SLC__1SDV_20250523T130238_20250523T130308_059322_075CCB_14B2
S1A_IW_SLC__1SDV_20250604T130237_20250604T130307_059497_0762D9_3AB5
S1A_IW_SLC__1SDV_20250616T130237_20250616T130306_059672_0768D1_F57D
S1A_IW_SLC__1SDV_20250628T130236_20250628T130306_059847_076EE3_6BFD

3. Differential SAR Interferometry (D-InSAR) for Displacement Measurement

D-InSAR analysis was performed to measure millimetre-scale deformation over time:



VISUALIZATIONS

1. Defining specific bridge location & parameters for satellite SAR based inspection



Project & Bridge Details

State – Rajasthan

Bridge No. – 899/1

Bridge Chainage – 974+880

Project - Chittorgarh to Kota

Zone – RJ 06

Structure Type – MNB

Sar Parameters

Sensor – Sentinel-1 (C-Band)

Beam Mode – IW (Interferometric Wide Swath)

Path – 34

Frame – 519

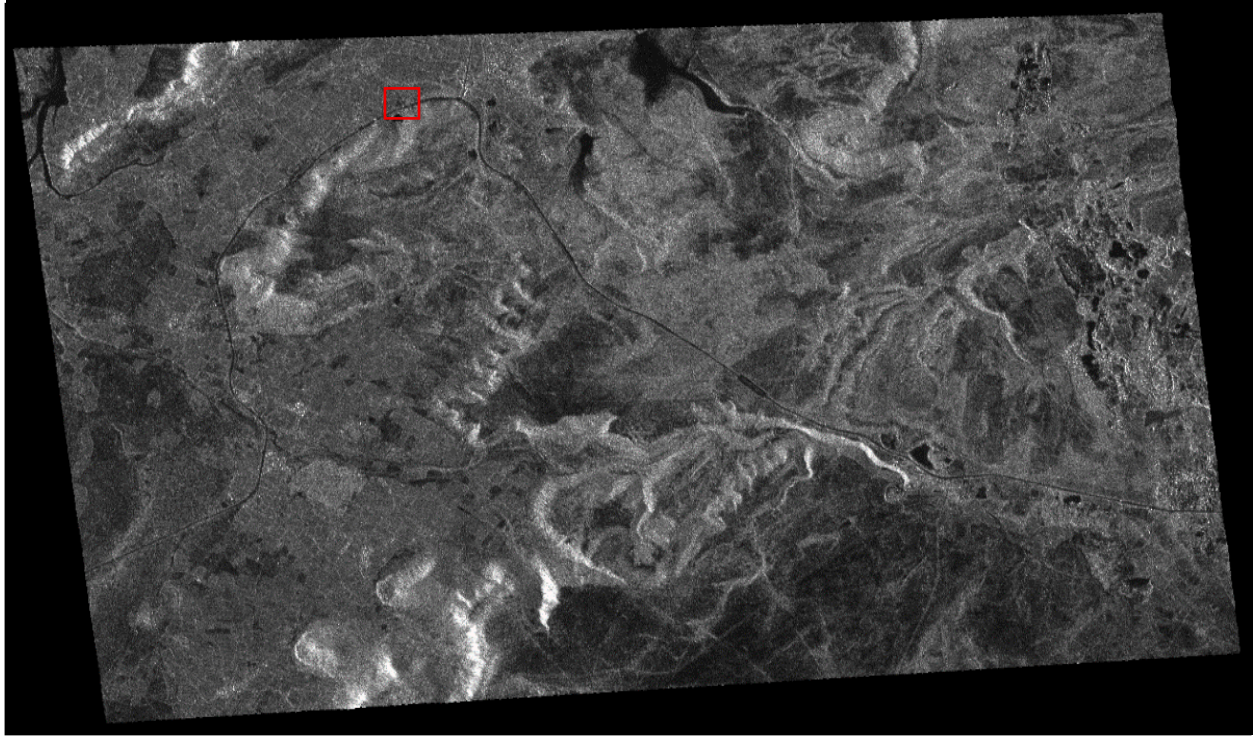
Flight Direction - DESCENDING

Polarization – VV+VH

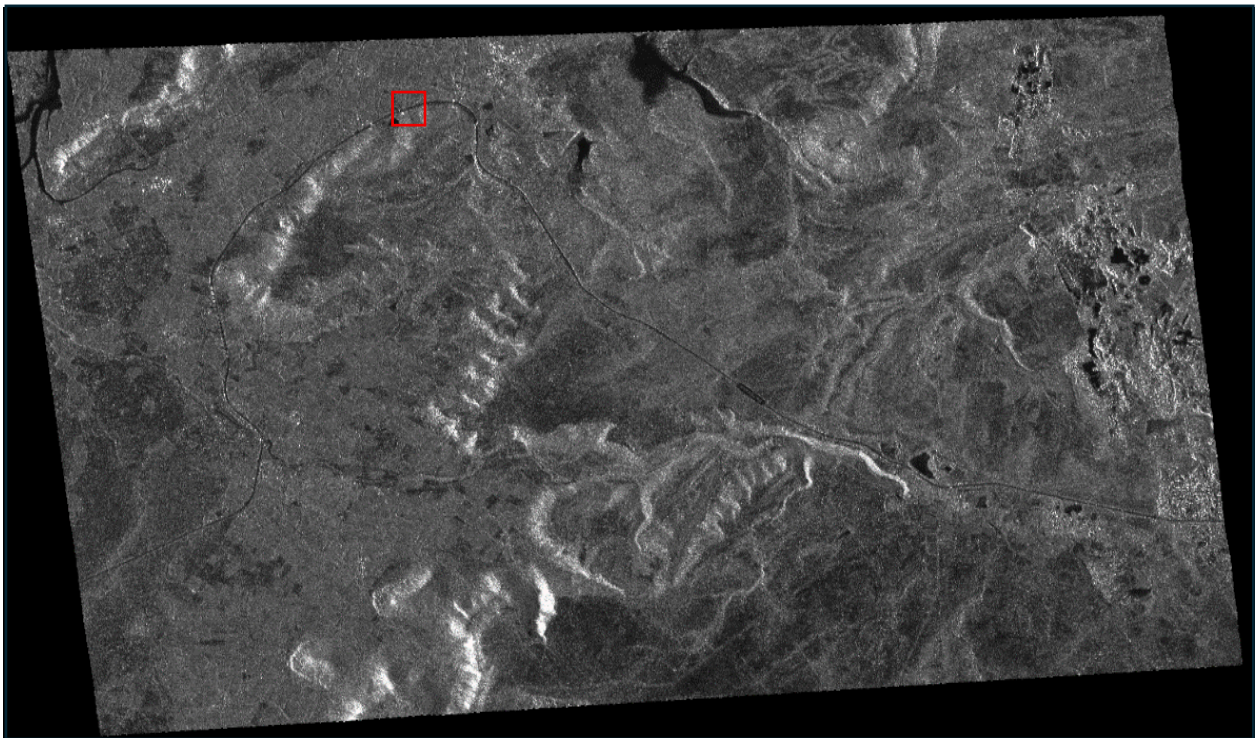
Absolute Orbit – 54706

2. Calibration, georeferencing, orthorectification of SAR images (VH/VV)

- Orthorectification of SAR image VH Polarization



- Orthorectification of SAR image VV Polarization

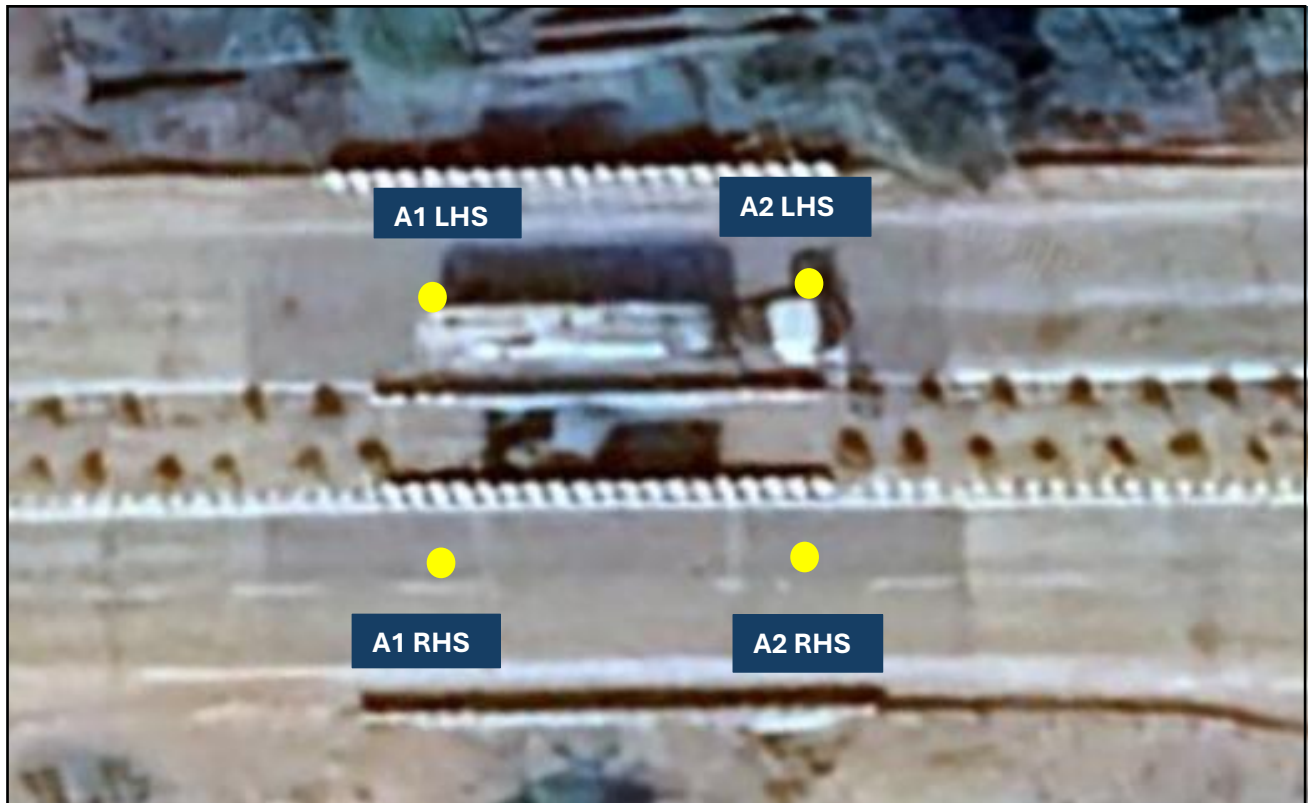


3. Results of SAR based bridge inspection, including visualizations

Bridge Components

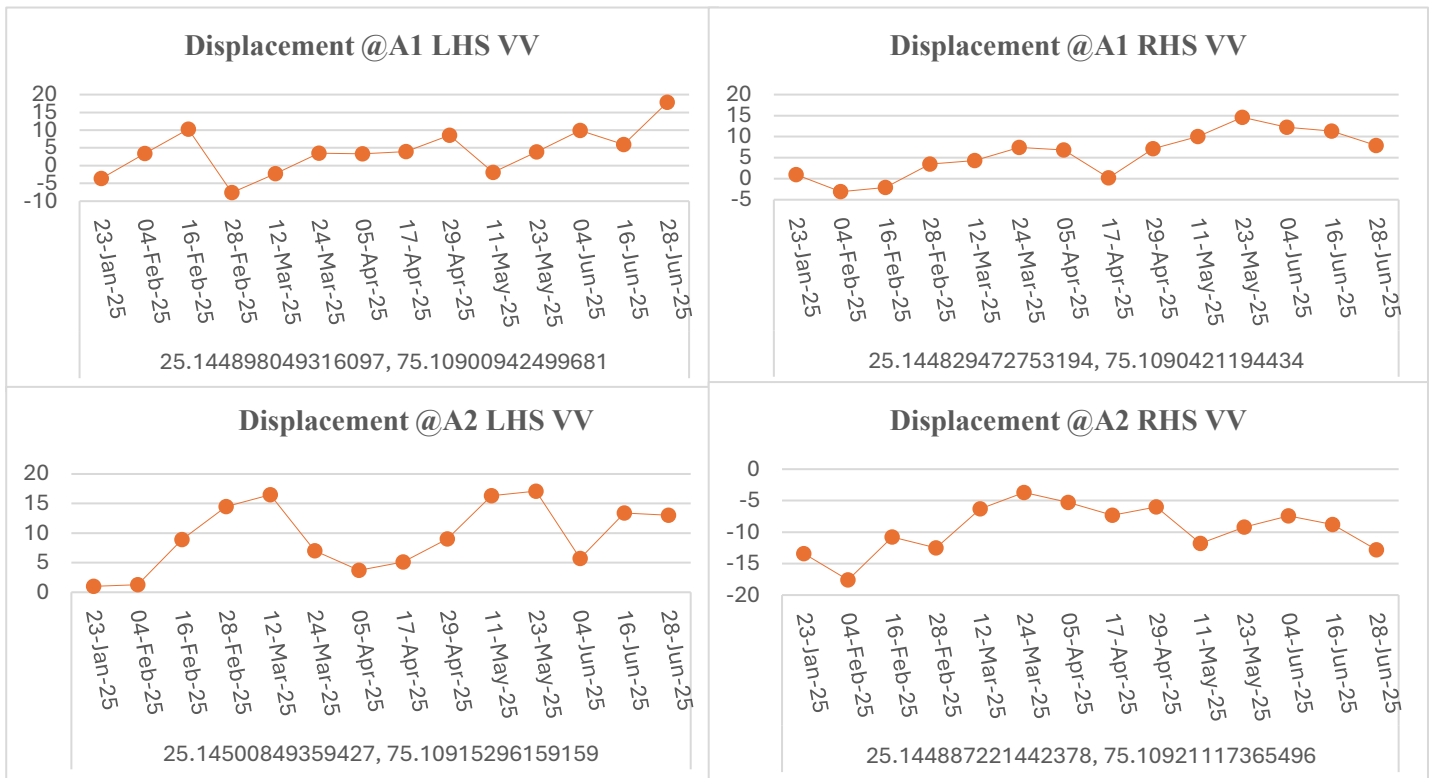
MNB – CH-974+880

Length – 23.86M

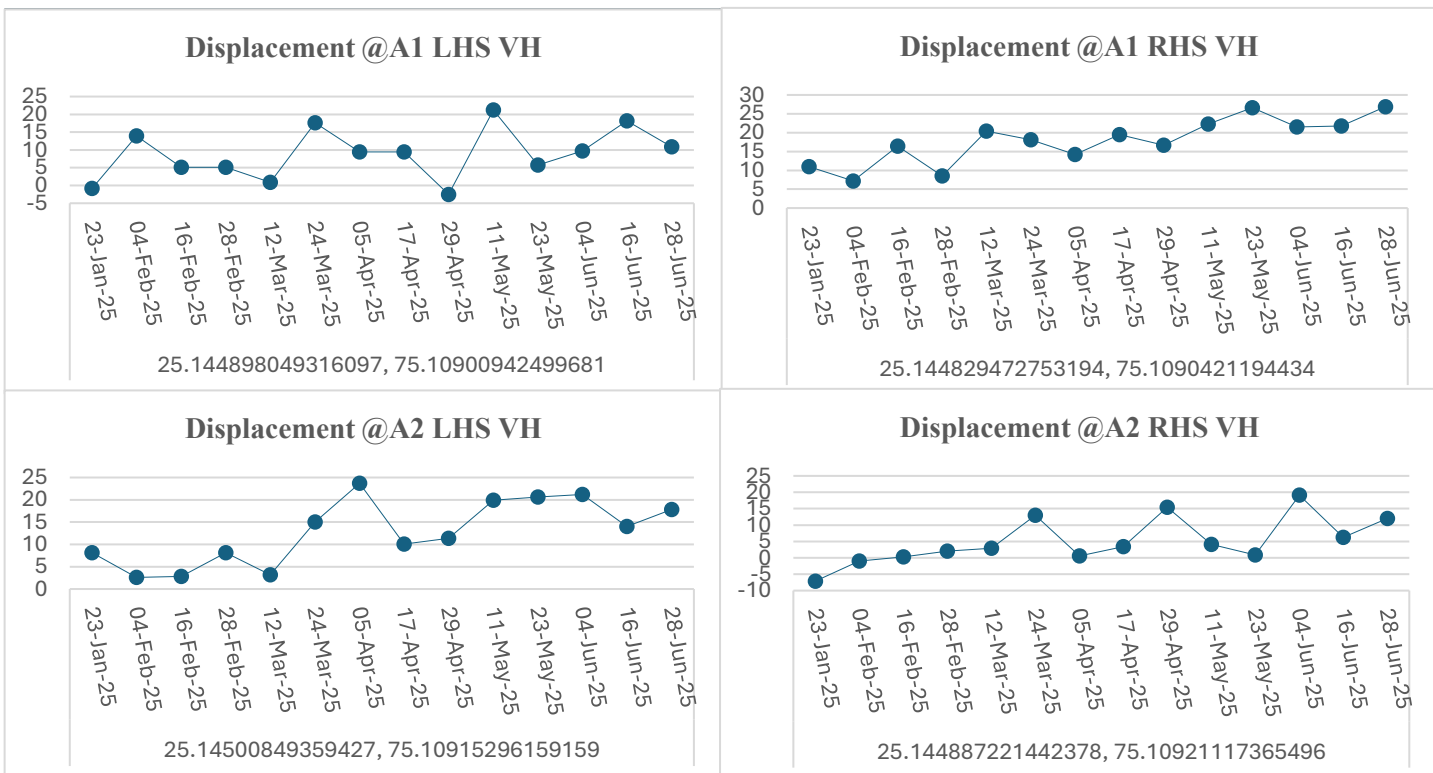


Scan Duration		From : 23 Jan 2025	To : 28 June 2025
Components & Abbreviation			
A1	Abutment 1		
A2	Abutment 2		
LHS	Left Hand Side		
RHS	Right Hand Side		

3.1 VV Displacement From 23 Jan 2025 To 28 June 2025



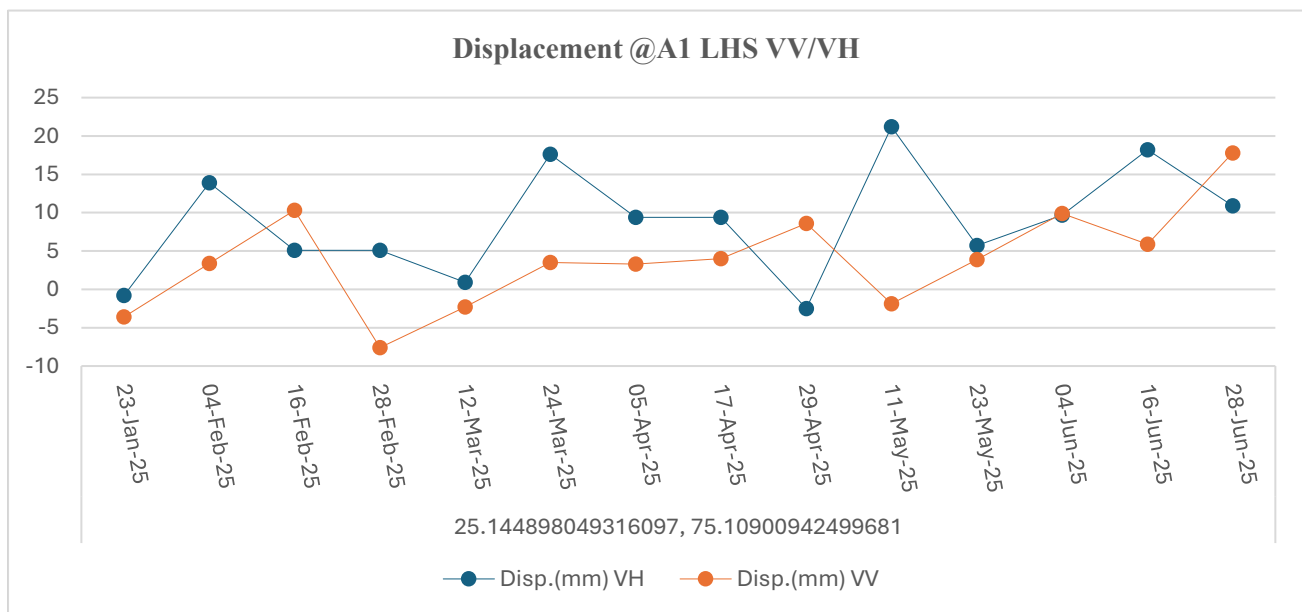
3.2 VH Displacement from 23 Jan 2025 To 28 June 2025



3.3 SAR Based Analysis at Abutment 1 LHS:



A1 LHS (25.144898049316097, 75.10900942499681)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** Vertical displacement exhibits moderate variability with both settlement and uplift phases over the monitoring period (Jan–Jun 2025).
- **Horizontal displacement (VH)** Horizontal displacement shows high variability with pronounced oscillations, ranging roughly between –3 mm and +21 mm.

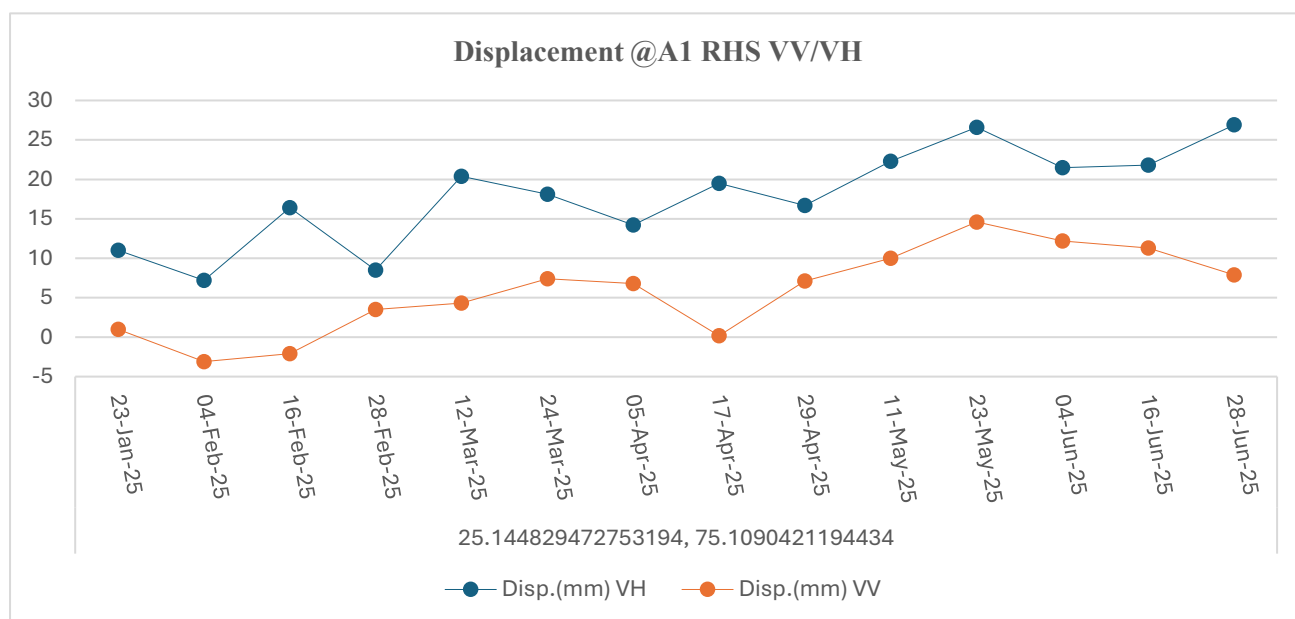
Critical Events & Anomalies:

- **February 2025:** VV records a sharp negative dip (–7 mm), while VH remains positive, indicating a temporary vertical settlement event without corresponding lateral reduction.
- **May 2025:** VH reaches one of its maximum values (+21 mm), coinciding with a short-term reduction in VV, suggesting a dominant lateral response during this period.
- **June 2025:** Both VV and VH rise simultaneously (VV +17 mm, VH +18–20 mm), indicating combined vertical uplift and lateral displacement intensification toward the end of the monitoring period.

3.4 SAR Based Analysis at Abutment 1 RHS:



A1 RHS (25.144829472753194, 75.1090421194434)



Key Observations:

Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** Horizontal displacement remains consistently positive and dominant, with values ranging approximately between +7 mm and +27 mm.
- **Vertical displacement (VV)** Vertical displacement shows an initial slight settlement during January–early February 2025, with VV values dipping to approximately –4 mm.

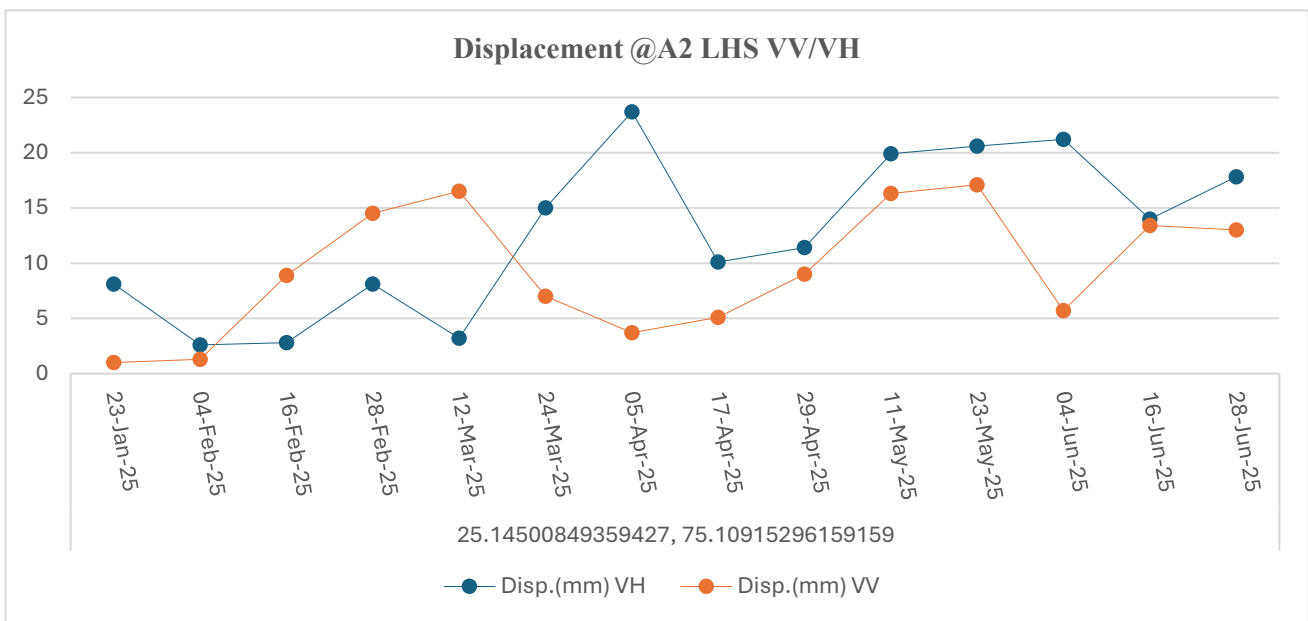
Critical Events & Anomalies:

- **February 2025:** A notable rise in VH to around +16–17 mm marks the onset of sustained lateral displacement growth, while VV begins transitioning from settlement to uplift.
- **April 2025:** VV briefly drops close to 0 mm, contrasting with elevated VH values (+18–20 mm), indicating a short-term decoupling between vertical and horizontal responses.

3.5 SAR Based Analysis at Abutment 2 LHS:



A2 LHS (25.14500849359427, 75.10915296159159)



Key Observations:

Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** Horizontal displacement exhibits significant variability with pronounced peaks, generally ranging between +3 mm and +24 mm.
- **Vertical displacement (VV)** Vertical displacement shows moderate to high variability, remaining predominantly positive throughout the monitoring period (Jan–Jun 2025).

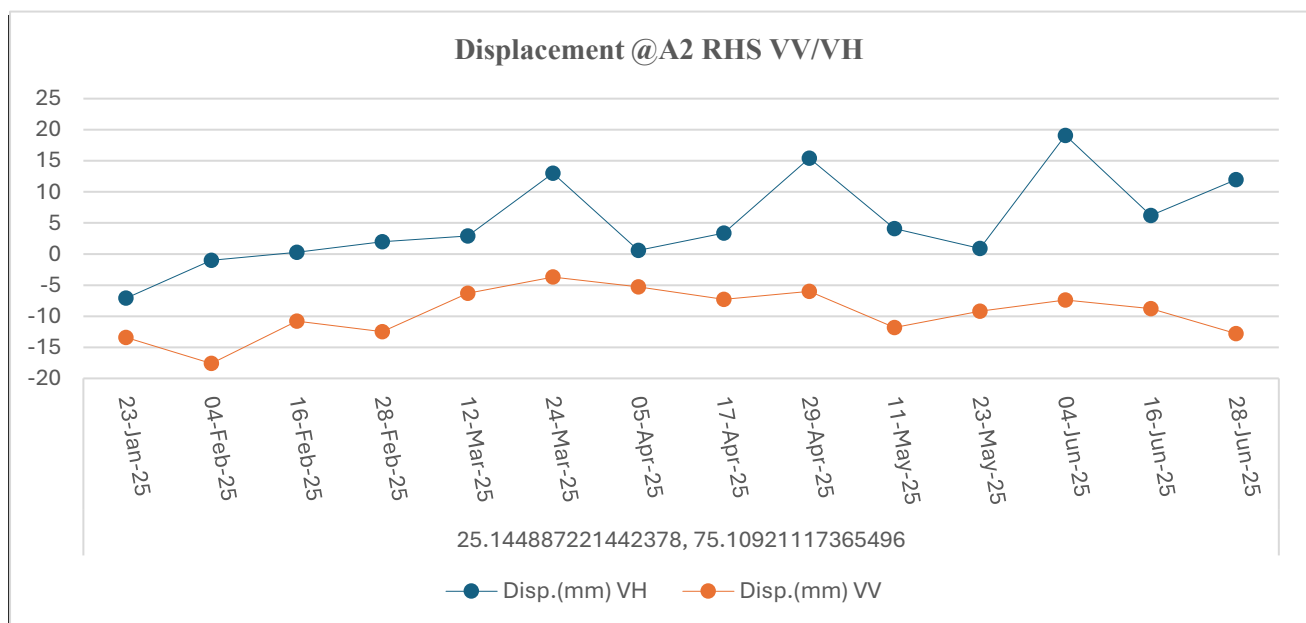
Critical Events & Anomalies:

- **February 2025:** VV rises rapidly to approximately +14–16 mm, while VH remains moderate, suggesting a period dominated by vertical uplift.
- **June 2025:** Both VV and VH maintain simultaneously elevated values (VV +15–17 mm, VH +20–22 mm), reflecting combined vertical and lateral movement intensification.

3.6 SAR Based Analysis at Abutment 2 RHS:



A2 RHS (25.144887221442378, 75.10921117365496)



Key Observations:

Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** Horizontal displacement shows high variability with repeated sharp peaks, generally ranging from approximately -7 mm to $+19$ mm.
- **Vertical displacement (VV)** Vertical displacement remains consistently negative throughout the monitoring period, indicating sustained downward movement. Values typically range between -5 mm and -17 mm, with moderate fluctuations over time.

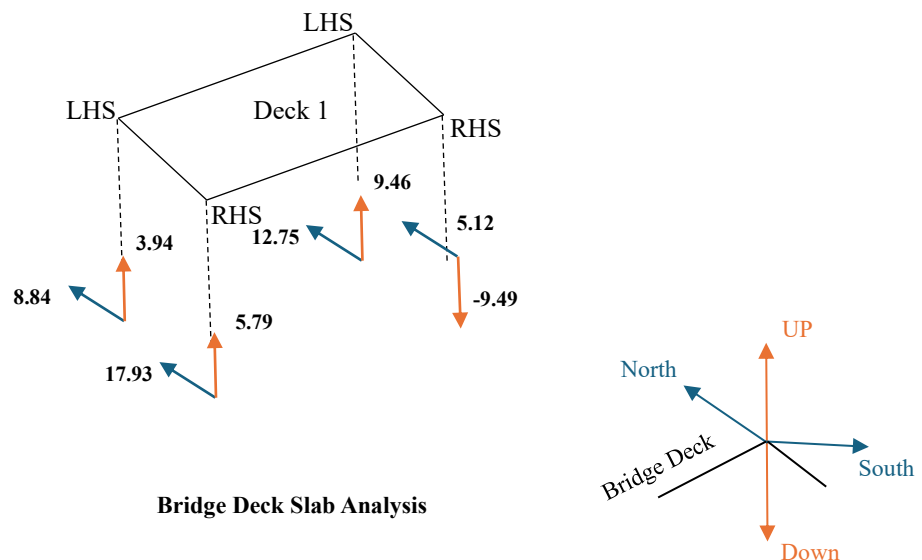
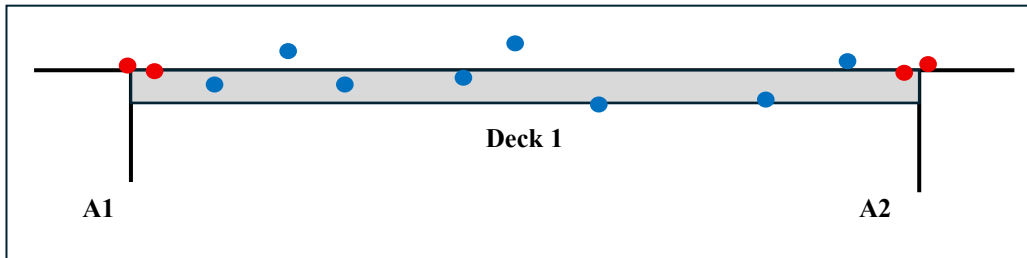
Critical Events & Anomalies:

- **March-April 2025:** VH experiences a sudden peak ($+14$ – 15 mm) while VV remains moderately negative (around -6 to -8 mm), suggesting dominant horizontal movement with limited vertical change.
- **June 2025:** VH remains elevated ($+12$ mm) while VV declines further (-14 mm), suggesting continued subsidence despite reduced lateral motion.

ANALYSIS

1. Identify potential deformation, shifts or movement of the bridge structure

- Points of Analysis (POA) of bridge (●)



Vertical Displacement Behaviour:

- Vertical displacements across Deck 1 show both uplift and downward movement, ranging approximately from +8.84 mm (uplift) to -9.49 mm.
- The left-hand side (LHS) exhibits net uplift, with vertical movements of approximately +3.94 mm to +5.79 mm, indicating upward response at the LHS supports.

Horizontal Displacement Behaviour:

- Horizontal displacements range approximately between 5.12 mm and 17.93 mm, with movement directions predominantly toward the north and northwest, as indicated by the vector orientation. The largest horizontal displacement 17.93 mm) is observed at the LHS lower monitoring point, while moderate horizontal movements of 12.75 mm and 8.84 mm are observed near the central and RHS locations

CONCLUSION:

- The displacement pattern for Deck 1 indicates a combination of moderate-to-high lateral movement and localized vertical differential displacement between the LHS and RHS. Overall behaviour appears structurally manageable, but the observed differential movements indicate a need for trend evaluation over time rather than immediate concern.